



Ivan Lopez-Mata

Date of birth: 20/06/2002 | **Place of birth:** Cunduacan, Tabasco, Mexico | **Nationality:** Mexican |
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ABOUT ME

Double-degree M.Sc. student in Chemistry, specializing in Physical Chemistry and Chemoinformatics through the Erasmus Mundus Chemoinformatics+ programme. Interested in computational chemistry, molecular modelling, machine learning, and data-driven molecular discovery. Experienced in electronic structure calculations, virtual screening workflows, molecular descriptors, and scientific computing environments, including Linux-based systems and workload management tools.

EDUCATION AND TRAINING

15/09/2025 – CURRENT Strasbourg, France

MASTER OF SCIENCE IN CHEMISTRY (CHEMOINFORMATICS) Université Strasbourg

Website <https://chimie.unistra.fr/formation/master/>

15/09/2025 – CURRENT Milan, Italy

MASTER OF SCIENCE IN CHEMISTRY (PHYSICAL CHEMISTRY) Università degli Studi di Milano

Website <https://www.unimi.it/it/corsi/laurea-magistrale/scienze-chimiche>

21/09/2020 – 05/09/2025 Cunduacan, Tabasco, Mexico

BACHELOR OF SCIENCE IN CHEMISTRY Juarez Autonomous University of Tabasco

Website <https://www.ujat.mx> | **Final grade** 9.64 | **Number of credits** 288

WORK EXPERIENCE

DATA SCIENCE INTERN – SIA SOFTWARE – 07/10/2024 – 24/01/2025 – VILLAHERMOSA, TABASCO, MEXICO

Website: <https://siasw.com/index.php/en/>

- Design and train artificial neural networks using Python to predict disease onset based on patients' clinical data.
- Develop the front end and back end of a Hospital Information System simulator using JavaScript and Flask, integrating the trained algorithms.
- Co-author an article detailing the obtained results from the project.

RESEARCH INTERN – CLUSTER OF EXCELLENCE RESOLV - RUHR UNIVERSITY BOCHUM – 14/05/2024 – 10/08/2024 – BOCHUM, GERMANY

Department: Center for Theoretical Chemistry | **Website:** <https://www.theochem.ruhr-uni-bochum.de>

- Use CP2K for Density Functional Theory calculations to obtain convergence of cutoff energy and lattice parameters.
- Use CPMD for ab initio molecular dynamics to study the Pt(111)/water interface in a basic solution.
- Write reports and present them to the research group.

RESEARCH VOLUNTEER – CHEMISTRY INSTITUTE - NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO – 13/09/2023 – 03/05/2024 – MERIDA, YUCATAN, MEXICO

- Develop and train Machine Learning models (SVM, KNN, Random Forest, Gradient Boosting Trees) to predict the biological activity of molecules using PubChem bioassays and molecular descriptors.
- Apply dimensionality reduction techniques (t-SNE, UMAP, PCA) to optimize data representation and evaluate model performance using key metrics such as Balanced Accuracy and F1-score.
- Co-author an article detailing the obtained results from the project.

IT TECHNICIAN – MOLECULAR MODELLING LABORATORY - JUAREZ AUTONOMOUS UNIVERSITY OF TABASCO – 13/02/2023 – 05/09/2025 – CUNDUACAN, TABASCO, MEXICO

- Perform preventive maintenance on the computer cluster.
- Reinstall the entire cluster's operating system and compile necessary programs for computational materials science.
- Write a manual on initial computer cluster configuration for computational materials science.

PUBLICATIONS

2026

[Setting up computational materials science clusters](#)

Practical and accesible guide for students, technicians and researchers looking to build a computer cluster from scratch, specifically geared towards the need of computational materials science.

Authors: Ivan Lopez Mata

2025

[Aplicación de modelos de inteligencia artificial para optimizar procesos de acoplamiento molecular en bases de datos de productos naturales](#)

Undergraduate graduation project completed under the case-study resolution modality, titled "Application of Artificial Intelligence Models to Optimize Molecular Docking Processes in Natural Product Databases." The project evaluated active learning strategies to improve large-scale virtual screening using the COCONUT natural products database, combining molecular fingerprints/descriptors with machine learning models to predict docking scores and efficiently enrich high-scoring ligands while reducing computational cost.

Authors: Ivan Lopez Mata, José M. Mora Fonz (supervisor)

2025

[Detection of Type 2 Diabetes Mellitus in individuals with normal fasting glucose using a multilayer perceptron neural network](#)

Full-paper accepted for presentation at the "2025 World Conference on Information Systems and Technologies".

Rubén Alfredo Peraza-Torres, Iván López-Mata, Alejandro Medina-Santiago, Ivonne Adriana Hernández-Hernández, José Antonio Cortes-García, Jorge Antonio Orozco-Torres.

2024

[De novo design of inhibitors of DNA methyltransferase 1: A critical comparison of ligand- and structure-based approaches](#)

Peer-reviewed journal article published at the Journal "Biomolecules".

Diana L. Prado-Romero, Fernanda I. Saldívar-González, Iván López-Mata, Pedro A. Laurel-García, Norberto Sánchez-Cruz, José L. Medina-Franco.

SOCIAL AND POLITICAL ACTIVITIES

01/11/2024 – 05/09/2025 Cunduacan, Tabasco

President of the Student Council

- Responsible for the meal and stationery support scholarship program.
- Record and analysis of student concerns to generate proposals for the university board.

01/11/2023 – 30/10/2024 Cunduacan, Tabasco

President of the American Chemical Society Student Chapter

- Design and maintenance of the student chapter's website, facilitating user account creation for professors and displaying an academic staff directory.
- Collect and disseminate opportunities (scholarships, exchanges, competitions) to enhance our students' scientific careers.
- Engage with national and international institutions, professors, and researchers to collaborate and bring diverse perspectives to our community.

SKILLS

Specialized Chemistry Software

CP2K | Density Functional Theory (VASP) | Quantum Chemistry software packages (Gaussian, ORCA, Multiwfn) | VMD, NAMD/ Gromacs (visualization, simulation of molecular systems) | Maestro-Schrodinger | Quantum Espresso

Data Science

Python DSci/ML Libraries (Numpy, Pandas, Sklearn, Scipy) | R, R Studio, R Markdown | RDKit

Web Development

Basic knowledge of Django and Flask | Begginer JavaScript for WebDevelopment

General Tools

Exprience using Latex | Linux servers (CentOS, Kali Linux, Devuan, Ubuntu) | Queuing systems: SLURM, MOAB/Torque | Linux (commands and shell scripting)

● LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	B2	B2	B2
ITALIAN	B2	B2	B1	B1	B1
FRENCH	A2	A2	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user